

MATHEMATICS

Max Marks: 20

TIME: 1hr

Instructions:

- All questions are compulsory.
- Answer all sections in the given sequence.
- Show all necessary working for calculations.

Section A:

Very Short Answer (1 marks each, 4 marks total)

Q.1 Which of the following rational numbers is in standard form?

(a) $\frac{20}{30}$

(b) $-\frac{1}{3}$

Q.2 Find the value of the following expressions at $a = 1$ and $b = -2$:

(a) $a^2 + b^2 + 3ab$

(b) $a^3 + a^2b + ab^2 + b^3$

Q.3 What are the coefficients of x in the following expressions?

(a) $4x - 3y$

(b) $y^2x - y$

Q.4 List three rational numbers between -2 and -1 .

Section B:
Short Answer (2 marks each, 16 marks total)

Q.1 Add the following expressions:

(a) $x^3 - x^2y - xy^2 - y^3$ and $x^3 - 2x^2y + 3xy^2 + 4y$

(b) $p^2qr + pq^2r + pqr^2$ and $-3pq^2r - 2pqr^2$

Q.2 Simplify the expression by combining the like terms:

(a) $7x^3 - 3x^2y + xy^2 + x^2y - y^3$

Q.3 What should be added to $3pq + 5p^2q^2 + p^3$ to get $p^3 + 2p^2q^2 + 4pq$?

Q.4 Solve the following:

(a) Find the sum of $-4\frac{5}{6}$ and $-7\frac{3}{4}$.

(b) Find the product of $-2\frac{3}{4}$ and $5\frac{6}{7}$.

Q.5 Draw the number line and represent the following rational numbers on it:

(a) $\frac{3}{4}$

(b) $\frac{-5}{8}$

(c) $\frac{-7}{4}$

(d) $\frac{7}{8}$

Q.6 Fill in the boxes with the correct symbol out of >, <, and =

(i) $\frac{-5}{7} \boxed{} \frac{2}{3}$

(ii) $\frac{-4}{5} \boxed{} \frac{-5}{7}$

(iii) $\frac{-7}{8} \boxed{} \frac{14}{-16}$

$\frac{1}{-3} \boxed{} \frac{-1}{4}$

Q.7 Solve :

(a) Simplify the expression and find its value when $a = 5$ and $b = -3$.

$$2(a^2 + ab) + 3 - ab$$

(b) If $z = 10$, find the value of $z^3 - 3(z - 10)$

Q.8 Match the following:

Column I

(i) $\frac{a}{b} \div \frac{a}{b}$

(ii) $\frac{a}{b} \div \frac{c}{d}$

(iii) $\frac{a}{b} \div (-1)$

(iv) $\frac{a}{b} \div \frac{-a}{b}$

(v) $\frac{b}{a} \div \left(\frac{d}{c}\right)$

Column II

(a) $\frac{-a}{b}$

(b) -1

(c) 1

(d) $\frac{bc}{ad}$

(e) $\frac{ad}{bc}$